# Software Requirements Specification for FitFolio: subtitle describing software

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# Revision History

| Date | Version | Notes |
|---|---|---|
| Date 1 | 1.0 | Notes |
| Date 2 | 1.1 | Notes |

[All capstone SRS documents should have a reflection appendix. —TPLT]

[This SRS template is based on the IEEE template, with input from co-pilot and S. Smith. It will get you started. You should not modify the section headings, without first discussing the change with the course instructor. Modification means you are not following the template, which loses some of the advantage of a template, especially standardization. —TPLT]

[Feel free to change the appearance of the report by modifying the LaTeX commands. —TPLT]

[This template document assumes that a single program is being documented. If you are documenting a family of models, you should start with a commonality analysis. —TPLT]

[The SRS is not generally written, or read, sequentially. The SRS is a reference document. It is generally read in an ad hoc order, as the need arises. —TPLT]

[Guiding principles for the SRS document:

- Do not repeat the same information at the same abstraction level. If information is repeated, the repetition should be at a different abstraction level.

—TPLT]

[The template description comments should be disabled before submitting this document for grading. —TPLT]

[You can borrow any wording from the text given in the template. It is part of the template, and not considered an instance of academic integrity. Of course, you need to cite the source of the template. —TPLT]

[When the documentation is done, it should be possible to trace back to the source of every piece of information. Some information will come from external sources, like terminology. —TPLT]

[An SRS document should have the following qualities: unambiguous, consistent, complete, validatable, abstract and traceable. —TPLT]

[The overall goal of the SRS is that someone can learn, understand and verify the captured domain knowledge. They should not have to trust the authors of the SRS on any statements. They should be able to independently verify/derive every statement made. —TPLT]

# 1 Introduction

[Explain what system is being specified. State purpose, users, and project goals. —TPLT]

[Introduce the name of your software as FitFolio. Using FitFolio as a command lets you change the name if necessary and have the change propagate to all documents. —TPLT]

[Include a reference to the template this work is based on. **?** justifies the SRS structure and **?** provides the broader life-cycle context that underlines modern requirements engineering practice. —TPLT]

## 1.1 Purpose

[State the purpose of the system. —TPLT]

[The purpose should be in the context of the project itself, not in the context of this course. Although the "purpose" of the document is to get a grade, you should not mention this. Instead, "fake it" as if this is a real project. —TPLT]

FitFolio is intended to solve ... [What problem does your program solve? The description here should be in the problem space, not the solution space. —TPLT]

## 1.2 Scope

[Briefly describe:

- Problem being solved
- Intended users
- Major functionality
- Expected benefits
- Clarifying the scope often means saying what the software doesn't do

—TPLT]

[The scope section is relevant for understanding the boundaries of the system. It is also relevant for later determining typical values of inputs. The scope should make it clear what inputs are reasonable to expect. This is a distinction between scope and context (context is a later section). Scope affects the inputs while context affects how the software will be used. —TPLT]

TODO

## 1.3 Stakeholders

- Customer
- End Users
- Project Sponsor
- Development Team
- Administrators

## 1.4 Definitions and Acronyms

[This section provides the meaning of the different words and phrases used in the domain of the problem. —TPLT]

---

| Term | Definition |
|---|---|
| TODO | TODO |

Table 1: Definitions and Acronyms

### 1.5 Characteristics of Intended Reader

[This section summarizes the skills and knowledge of the readers of the SRS. It does NOT have the same purpose as the "User Characteristics" section (Section 2.4). The intended readers are the people that will read, review and maintain the SRS. They are the people that will conceivably design the software that is intended to meet the requirements. The user, on the other hand, is the person that uses the software that is built. They may never read this SRS document. Of course, the same person could be a "user" and an "intended reader." —TPLT]

[The intended reader characteristics should be written as unambiguously and as specifically as possible. Rather than say, the user should have an understanding of physics, say what kind of physics and at what level. For instance, is high school physics adequate, or should the reader have had a graduate course on advanced quantum mechanics? —TPLT]

## 2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

[The purpose of this section is to provide general information about the system so that the specific requirements in the later sections will be easier to understand. The general system description section is designed to be changeable independent of changes to the functional requirements documented in the specific system description. The general system description provides a context for a family of related models. The general description can stay the same, while specific details are changed between family members. —TPLT]

### 2.1 Problem Statement

**Current situation.** Building good fitness and financial habits early pays off for decades, because both compound over time. Regular exercise builds on itself, and so does money invested early. Poor habits compound in the same way. Yet many young adults, particularly students and those starting their first jobs, struggle to begin either one.

**Problem.** Advice on managing money is scattered across many sources, which often contradict each other, and learning to invest by doing is risky because beginners' mistakes cost real money. Starting to exercise is similarly daunting, and many beginners lose motivation before they see results.

**Impact.** Many people put off both habits until later, when poor habits are already established and harder to change.

**Desired improvement.** An approachable, low-risk way for beginners to build fitness and financial habits together while staying motivated.

## 2.2 Product Vision

FitFolio is a mobile app in which users earn virtual currency from their everyday physical activity and invest it in a simulated market that follows real-world prices. It lets beginners build fitness habits and learn to invest without risking real money. Its goals, which the requirements in this document discuss, are:

**G1 (Learning by doing).** Users learn core financial and fitness concepts through hands-on, low-risk practice, with each decision paired with feedback explaining what happened and why. *Measured by:* a before-and-after knowledge check on a defined set of core concepts, where at least 70% of usability test participants improve their score.

**G2 (A human-centered, all-in-one experience).** Users see their fitness progress and financial learning together in one place, so they rarely need to leave the app to understand what they are looking at. *Measured by:* usability testing, where at least 80% of first-time participants complete the core tasks without outside help.

**G3 (Motivation through friends, with privacy).** Users stay motivated by sharing their progress with friends, while keeping control of their data. *Measured by:* users control who sees each shared statistic, and no raw health data is shared unless the user selects it.

**G4 (Risk-free investing practice).** Users practice investing in a simulated market based on real-world market data, without risking real money. *Measured by:* users can trade at least 50 real-world stocks at prices within 15 minutes of the real market, and can practice in market scenarios such as a replay of a past market downturn.

**G5 (Rewards for everyday activity).** Users are rewarded for the activities they already do, as recorded by their phone's health platform. *Measured by:* at least five activity types are supported.

## 2.3 System Context

Figure 1 shows FitFolio as a single system, with its mobile app, server and database inside the boundary, and the external entities it exchanges information with. Activity sources, such as phone sensors, smartwatches and other tracker apps, record activity in the phone's health platform. They never communicate with FitFolio directly, so they are shown outside the boundary with a dashed outline. FitFolio does not connect to banks, brokerages, personal portfolio trackers, or the phone's location services or contacts. Reminders are generated on the phone as local notifications, so FitFolio uses no push notification service.

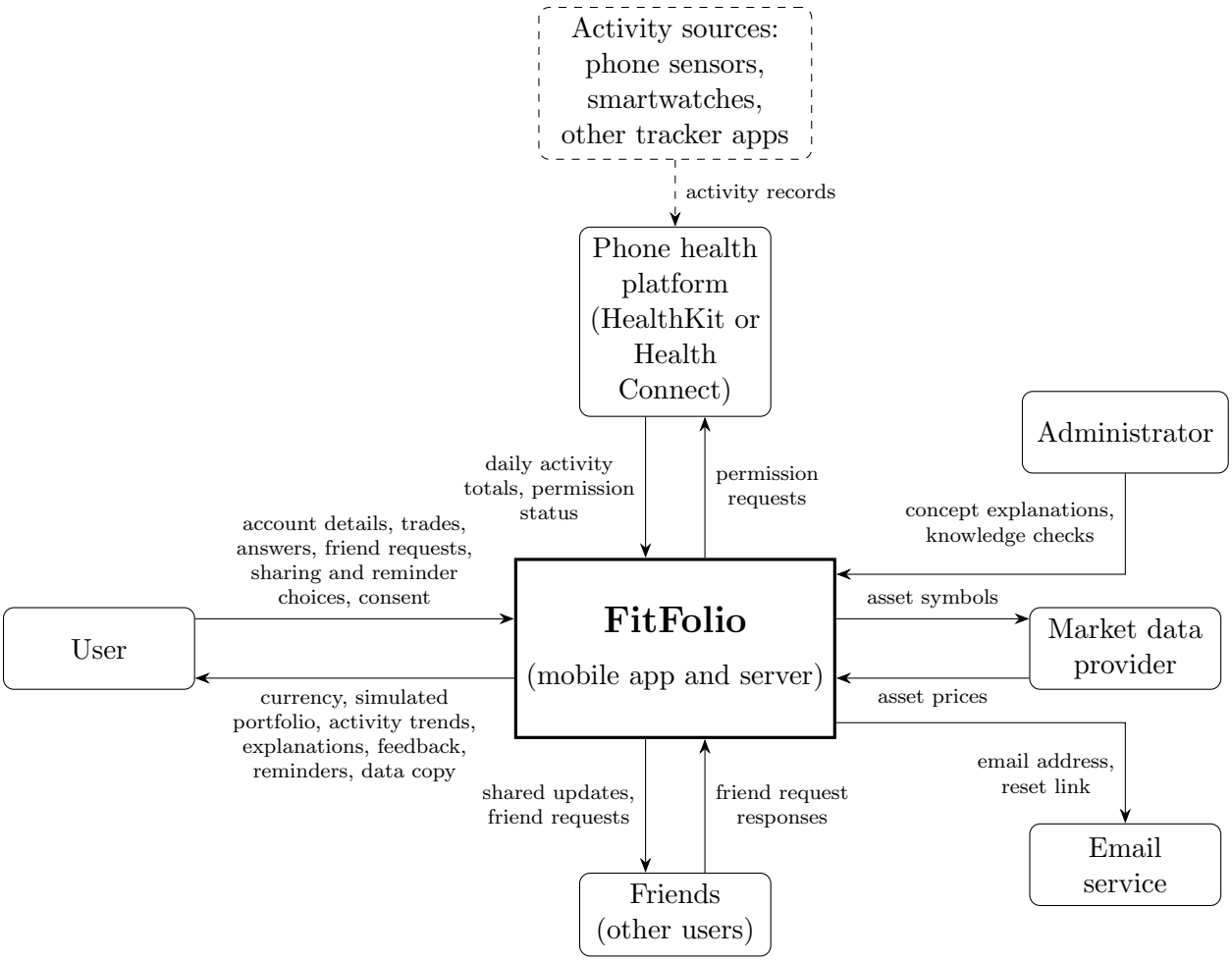



Figure 1: System context of FitFolio

The responsibilities of each entity in Figure 1 are:

- User Responsibilities:
  - Confirm truthfully that they are at least `MIN_USER_AGE` years old.
  - Keep any profile measurements they choose to enter, such as height and weight, accurate.
  - Grant FitFolio the permissions its features use, and keep an activity source recording to the phone's health platform.
  - Make their own trade decisions. FitFolio gives no financial advice.
  - Choose what to share, with which friends, and when to receive reminders.
  - Keep an Internet connection for trading and the social feed.
  - Keep their password private and keep access to their account's email address.

- FitFolio Responsibilities:
  - Reject profile measurements outside plausible ranges.
  - Award currency only for activity within plausibility limits and up to a daily cap (Hazard Analysis SAF-7, SEC-1).
  - Treat a day with no activity data as a day with zero activity.
  - Reject purchases that cost more than the user's balance and sales of more shares than the user holds (Hazard Analysis SEC-5).
  - Record each trade once and completely (Hazard Analysis SEC-5, SEC-6).
  - Show each asset price with the time it was obtained, and mark delayed prices (Hazard Analysis SEC-2).
  - Refuse trades on out-of-date prices while the market is open, and queue trades requested while it is closed (Hazard Analysis SEC-3, SEC-4).
  - Show each shared update only to the friends the user chose, without location detail (Hazard Analysis SEC-13, SEC-14).
  - Request only the permissions its features use, explaining each one (Hazard Analysis SEC-7, SEC-8).
  - Send the market data provider only the symbols of the assets being priced, and no personal or health data (Hazard Analysis SEC-11).
  - Send the email service no personal or health data other than the user's email address and their password reset link (Hazard Analysis SEC-11).
- Health Platform Responsibilities:
  - Store activity records from the user's activity sources, and provide daily totals for the activity types the user has permitted.
  - Grant or withdraw FitFolio's read access as the user chooses, and report the current permission status.
  - Keep the activity data it records as accurate as its activity sources allow.
- Market Data Provider Responsibilities:
  - Supply current and historical prices for the assets FitFolio requests, within its request limits.
- Email Service Responsibilities:
  - Deliver each password reset link that FitFolio sends to the user's email address.
- Administrator Responsibilities:

  - Keep concept explanations and knowledge checks accurate, and name no asset for the user to buy or sell in them (Hazard Analysis SAF-9).

- Friend Responsibilities:

  - Accept or decline a user's friend request.
  - View the updates a user has shared with them in their social feed.

FitFolio is intended for personal education and habit building. Members of the general public use it on their own smartphones to track their activity, earn currency and practice investing in a simulated market. It is not intended for professional, engineering or scientific use.

FitFolio is not mission-critical: it holds and moves no real money, so an outage only delays rewards and learning. It is not safety-critical either since it controls no physical equipment and provides no medical function. However, it handles personal health data and is designed to influence how users exercise and make financial decisions. These hazards, and the requirements that control them, are covered in the Hazard Analysis. The rationale in Section 11 therefore focuses on those requirements.

## 2.4 User Characteristics

The typical user of FitFolio is a young adult, such as a student or someone starting their first job (Problem Statement and Goals, Section 1). The user of FitFolio:

- Is at least `MIN_USER_AGE` years old (Hazard Analysis CA2).
- Uses a smartphone daily, and can install an app, grant or deny app permissions, and use common mobile app controls.
- Carries an Android or iOS phone during everyday activity. No wearable device or fitness expertise is required.
- Reads English, and has the numeracy of Grade 9 mathematics: working with percentages, comparing values, and reading a simple line chart.
- Needs no prior knowledge of investing terms such as stock, exchange-traded fund (ETF) or portfolio.

As per interviews conducted, participants mainly track their activity with a phone. Both already knew some investing concepts, through self-study or formal study, and both learned most by doing. No one interviewed was a self-reported complete beginner, so the beginner characteristics above come from the Problem Statement, and usability testing should include beginners.

## 2.5 System Constraints

The following constraints limit the design of FitFolio. Each one is imposed by the project brief, an external service, the law or the app stores.

SC1: FitFolio shall run as a mobile app on Android phones running Android `MIN_ANDROID_VERSION` or later and on iPhones running iOS `MIN_IOS_VERSION` or later. *Rationale:* This project calls for a mobile app, and these are the versions on the devices available to the team and the supervisor for testing (Development Plan, Section 10).

SC2: FitFolio shall obtain recorded activity only through the phone's health platform: Apple HealthKit on iOS and Health Connect on Android. This rule also covers any library FitFolio uses to read health data. No core feature shall require a wearable device. *Rationale:* Users carry their phone during everyday activity (Section 2.4), and the health platforms already collect data from the phone and from any wearable the user owns (note, wearable integration is a stretch goal).

SC3: FitFolio shall access health data only under the access rules Apple (iOS provider) and Google (Android OS provider) set for their health platforms, such as declaring each data type it reads and explaining each permission to the user (**??**). *Rationale:* Apps that break these rules are not accepted in the app stores. The rules also limit what FitFolio can read, such as how far back past activity data can be read.

SC4: FitFolio shall obtain real-world asset prices from a third-party market data provider (`MARKET_DATA_PROVIDER`), within the provider's request limits and licence terms. *Rationale:* The team cannot produce market prices itself, so request limits, cost and permitted use are set by the provider.

SC5: Virtual currency and simulated holdings shall have no monetary value. They cannot be bought with, redeemed for or exchanged for money, and FitFolio shall not connect to any bank or brokerage account or give personalized investment advice. FitFolio shall state the associated risks whenever it explains a financial concept. *Rationale:* Handling real money or giving personalized investment advice would bring FitFolio under financial regulation that the team is not licensed for. Keeping real money out also keeps the practice risk-free (G4).

SC6: FitFolio shall collect, use, store and delete personal and health data in line with Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) and the health-data policies of the Apple App Store and Google Play. *Rationale:* These laws and policies apply to any app that handles this data in Canada and in these stores. The specific controls are in Section 7.3 and the Hazard Analysis (SEC-7 to SEC-12 and SEC-18).

SC7: Trading, market prices and the social feed shall require an Internet connection to FitFolio's server. *Rationale:* Prices are fetched from the provider once for all users to

stay within its request limits (SC4), friends' shared updates must reach other users' phones, and a user's portfolio must survive the loss of their phone.

# 3 Stakeholder Requirements

[Technology-independent requirements. These come from statements made by the stakeholders. You will find these through interviews and surveys. They are often expressed in natural language. They are not necessarily testable, but they should be verifiable. —TPLT]

[The stakeholder requirements are numbered so that you can have traceability to other parts of your document. —TPLT]

SR1 TODO

SR2 TODO

SR3 TODO

[You should give your SRs meaningful labels, so that you can refer to them in other parts of the SRS. The label names should be meaningful mnemonics, not numbers. —TPLT]

# 4 User Personas

[Create 2-5 realistic personas. Personas are fictional characters that represent the different user types that might use your software. They are used to help understand the users' needs, experiences, behaviors and goals. They are not real people, but they should be based on real data collected from user research. —TPLT]

## 4.1 Persona 1

- Name:
- Description:
- Goals:
- Frustrations:
- Technical Ability:

## 4.2 Persona 2

- Name:
- Description:

- Goals:
- Frustrations:
- Technical Ability:

# 5 Use Cases

[Use cases are a way to describe the interactions between users and the system. They are typically written in natural language and describe the steps that a user takes to achieve a specific goal. Use cases can be used to identify functional requirements and to create test cases. —TPLT] [Typically 8-15 substantial use cases. —TPLT]

## 5.1 UC1: Example Use Case

**Primary Actor**

TODO

**Preconditions**

- TODO

**Main Success Scenario**

1. TODO
2. TODO
3. TODO

**Alternative Flows**

- TODO

**Postconditions**

- TODO

# 6 Functional Requirements

This section lists what FitFolio shall do, grouped by feature area. Each requirement has a fit criterion and traces to the stakeholder requirements and use cases it refines and to the goals in Section 2.2. A requirement that begins with a Hazard Analysis (HA) identifier, such as (HA SEC-3), appears in the Hazard Analysis with the same identifier and wording. A requirement whose traces end with a Hazard Analysis identifier, such as Hazard Analysis SEC-13, implements part of that Hazard Analysis requirement.

## 6.1 Accounts and Profile

FR1: The system shall let a new user create an account with an email address and a password.
*Fit criterion:* A test user who enters an unused email address and a valid password has an account, and can sign in with them.
*Traces to:* SR**??**, G2

FR2: The system shall not create an account until the user confirms that they are at least `MIN_USER_AGE` years old.
*Fit criterion:* A test user who does not confirm their age during sign-up has no account created.
*Traces to:* SR**??**, G3, Hazard Analysis CA2

FR3: The system shall require each user to choose a username that no other user has.
*Fit criterion:* A test user who tries to sign up with a username that another account already uses is asked to choose a different one.
*Traces to:* SR**??**, G3

FR4: The system shall let a user sign in with their email address and password.
*Fit criterion:* A test user who enters their correct email address and password is signed in; one who enters a wrong password is not.
*Traces to:* SR**??**, G2

FR5: The system shall let a signed-in user sign out.
*Fit criterion:* After a test user signs out, opening the app shows the sign-in screen and none of their data.
*Traces to:* SR**??**, G3

FR6: The system shall let a user who has forgotten their password set a new one through a link sent to their account's email address.
*Fit criterion:* A test user who requests a reset receives a link at their email address, and after setting a new password can sign in only with the new one.
*Traces to:* SR**??**, G2

FR7: The system shall let the user view and edit their profile details.
*Fit criterion:* A test user who changes a profile detail sees the new value the next time they open their profile.
*Traces to:* SR**??**, G3

FR8: The system shall not require the user to enter their height or weight in order to use any feature.
*Fit criterion:* A test user who leaves height and weight empty can earn currency, trade, learn, and share.
*Traces to:* SR**??**, SR**??**, G3, G5

FR9: The system shall let the user download a copy of the personal data FitFolio stores about them.
*Fit criterion:* A test user who requests a copy receives a file containing their profile, activity totals, balance, holdings, trades, and shared updates.
*Traces to:* SR??, G3

FR10: Before deleting an account, the system shall ask the user to confirm the deletion in a message stating that it cannot be undone.
*Fit criterion:* A test user who starts account deletion is shown this warning, and their account still exists if they cancel.
*Traces to:* SR??, UC??, G3

FR11: The system shall let the user delete their account from within the app.
*Fit criterion:* A test user who confirms deletion can no longer sign in, and none of their data is shown to them or to their friends.
*Traces to:* SR??, UC??, G3, Hazard Analysis SEC-18

## 6.2 Activity and Rewards

FR12: The system shall award virtual currency for at least `MIN_ACTIVITY_TYPES` activity types (the supported activity types).
*Fit criterion:* A tester records activity of `MIN_ACTIVITY_TYPES` different activity types in the health platform, and each one earns currency.
*Traces to:* SR??, UC??, G5

FR13: The system shall award virtual currency for recorded activity of a supported activity type.
*Fit criterion:* For a test user whose recorded activity in one calendar day is plausible and would earn at most `MAX_DAILY_CURRENCY`, the balance increases by the sum, over the supported activity types, of each type's daily total multiplied by its rate in `EARNING_RATES`.
*Traces to:* SR??, UC??, G5

FR14: The system shall treat a day with no activity data from the health platform as a day with zero activity.
*Fit criterion:* For a test user whose health platform reports no activity data for a day, that day shows zero activity and earns no currency.
*Traces to:* SR??, UC??, G5

FR15: (HA SEC-1) The system shall not count an activity record that exceeds the plausibility limits for its activity type toward the user's currency, activity trends, streak or milestones, and shall show the user that the record was not counted.
*Fit criterion:* A test record of more than `MAX_DAILY_STEPS` steps in one day, or a workout longer than `MAX_WORKOUT_DURATION`, earns no currency, does not appear in the

user's activity trends, streak or milestones, and is shown to the user as not counted.
*Traces to:* SR??, UC??, G5

FR16: (HA SAF-7) The system shall award a user no more than `MAX_DAILY_CURRENCY` virtual currency for recorded activity in any calendar day.
*Fit criterion:* For a test user whose recorded activity in one calendar day is plausible and would otherwise earn more than `MAX_DAILY_CURRENCY`, the balance increases by exactly `MAX_DAILY_CURRENCY` for that day. Each activity counts toward the calendar day in the user's time zone at the time of the activity.
*Traces to:* SR??, UC??, G5

FR17: The system shall let the user set or change a daily activity goal for a supported activity type, with each change applying from the next day.
*Fit criterion:* A test user who changes their daily activity goal keeps the previous goal for the rest of that day and has the new goal from the next day.
*Traces to:* SR??, G5

FR18: (HA SAF-3) The system shall not require the user to set an activity goal in order to earn currency for recorded activity.
*Fit criterion:* A test user with no activity goal set who records activity receives currency for it.
*Traces to:* SR??, UC??, G5

FR19: (HA SAF-4) The system shall not change a user's activity goal without the user's confirmation.
*Fit criterion:* A test user who exceeds their goal on `GOAL_REVIEW_PERIOD` consecutive days still has the same goal unless they confirm a change.
*Traces to:* SR??, G5

FR20: The system shall show the user their current streak.
*Fit criterion:* A test user with no streak who meets their activity goal on three consecutive days is shown a streak of 3. After a following rest day, they are shown a streak of 4. After a following day on which they neither meet their goal nor take a rest day, they are shown a streak of 0.
*Traces to:* SR??, G5

FR21: (HA SAF-1) The system shall let the user mark up to `REST_DAYS_PER_PERIOD` days in each `STREAK_PERIOD` as rest days, and a rest day shall not end the user's activity streak.
*Fit criterion:* A test user with a streak who marks a day as a rest day and records no activity that day does not have the streak reset to zero at the start of the following day. A request to mark more than `REST_DAYS_PER_PERIOD` rest days in one `STREAK_PERIOD` is refused.
*Traces to:* SR??, G5

FR22: (HA SAF-2) The system shall not reduce a user's virtual currency balance or simulated holdings when they miss an activity goal or their streak ends, and shall not show a missed goal or an ended streak to other users.
*Fit criterion:* For a test user who misses their goal on day *d*, the currency balance and simulated holdings at the start of day *d+1* equal those at the end of day *d*. No friend's social feed shows the missed goal or the ended streak.
*Traces to:* SR**??**, SR**??**, G3, G5

FR23: The system shall show the user each milestone in `MILESTONES` when they reach it.
*Fit criterion:* A test user whose streak, or total currency earned, first reaches a value in `MILESTONES` is shown that milestone once.
*Traces to:* SR**??**, G5

FR24: The system shall show the user their daily totals for each supported activity type over time.
*Fit criterion:* A test user with recorded activity on 14 days sees the total for each of those days, for each supported activity type.
*Traces to:* SR**??**, G2, G5

## 6.3 Simulated Market and Portfolio

FR25: The system shall offer at least `MIN_TRADABLE_ASSETS` assets for trading in the simulated market.
*Fit criterion:* A tester counts the assets that can be traded in the simulated market and finds at least `MIN_TRADABLE_ASSETS`.
*Traces to:* SR**??**, UC**??**, G4

FR26: The system shall price each asset in the simulated market at the most recent price it has obtained for that asset from the market data provider.
*Fit criterion:* For a test asset, the price FitFolio shows and uses for trades equals the latest price FitFolio received from the market data provider for that asset.
*Traces to:* SR**??**, UC**??**, G4

FR27: (HA SEC-2) The system shall display each asset price with the time it was obtained, and shall mark the price as delayed when it is older than `QUOTE_STALENESS_LIMIT`.
*Fit criterion:* A test price obtained more than `QUOTE_STALENESS_LIMIT` ago is shown with its time and a "Delayed" marker. A price obtained within the limit is shown with its time and no marker.
*Traces to:* SR**??**, UC**??**, G4

FR28: (HA SAF-8) Before the user's first trade, the system shall display a notice stating that the market is simulated, that virtual currency and holdings have no real-world value, and that nothing in FitFolio is financial advice. The system shall not complete the first trade until the user acknowledges the notice.

*Fit criterion:* A new test user who attempts their first trade is shown the notice and cannot complete the trade until they acknowledge it. The notice contains all three statements: the market is simulated, currency and holdings have no real-world value, and nothing in FitFolio is financial advice.
*Traces to:* SR??, UC??, G4

FR29: The system shall let the user buy shares of an asset using virtual currency.
*Fit criterion:* While a test asset's market is open, a test user with enough currency buys one share of it. Their balance decreases by the price shown when they confirmed the trade, and their holding of it increases by one.
*Traces to:* SR??, SR??, UC??, G4

FR30: The system shall let the user sell shares of an asset they hold for virtual currency.
*Fit criterion:* While a test asset's market is open, a test user holding one share of it sells it. Their balance increases by the price shown when they confirmed the trade, and their holding of it decreases by one.
*Traces to:* SR??, SR??, G4

FR31: (HA SEC-5) The system shall record each trade completely, updating both the currency balance and the holdings, or not at all. It shall reject a purchase that costs more than the user's currency balance and a sale of more shares than the user holds.
*Fit criterion:* A test trade interrupted before it completes leaves the balance and holdings unchanged, and a completed trade changes both. A purchase costing more than the balance, or a sale of more shares than are held, is rejected and changes neither.
*Traces to:* SR??, UC??, G4

FR32: (HA SEC-6) The system shall record a trade only once when the same trade request is submitted more than once.
*Fit criterion:* When a test user submits a trade and the same request is sent again, for example by retrying after a dropped connection, exactly one trade is recorded.
*Traces to:* SR??, UC??, G4

FR33: (HA SEC-3) The system shall not execute a trade while the asset's market is open and the asset's price is older than `QUOTE_STALENESS_LIMIT`.
*Fit criterion:* While a test asset's market is open and its price is older than `QUOTE_STALENESS_LIMIT`, a trade request for it is refused and the user is told the price is out of date. With a price within the limit, the trade proceeds.
*Traces to:* SR??, UC??, G4

FR34: (HA SEC-4) The system shall queue a trade requested while the asset's market is closed, and execute it at the first price obtained after the market next opens.
*Fit criterion:* A test trade requested while the market is closed is shown to the user as queued, with a notice that it will run at the opening price, and is executed at the first price obtained after the market opens.
*Traces to:* SR??, UC??, G4

FR35: The system shall show the user each of their holdings.
*Fit criterion:* A test user who owns shares of two assets sees both holdings, each with the number of shares held.
*Traces to:* SR??, UC??, G2, G4

FR36: The system shall show the user their portfolio value.
*Fit criterion:* For a test user with known holdings, the portfolio value shown equals the sum of each holding multiplied by its asset's most recent price.
*Traces to:* SR??, UC??, G2, G4

FR37: (HA SAF-10) The system shall label every displayed return and portfolio value as simulated.
*Fit criterion:* Every screen that shows a return or portfolio value shows a "Simulated" label next to it.
*Traces to:* SR??, UC??, G4

FR38: (HA SAF-9) The system shall not recommend that a user buy or sell a specific asset.
*Fit criterion:* No screen or notification shown to a test user tells them to buy or sell a named asset, or presents assets as a "recommended pick".
*Traces to:* SR??, UC??, G1, G4

## 6.4 Learning and Feedback

The requirements in this subsection refer to the core concepts in Table 2. These are the concepts measured by the knowledge check in goal G1; the list is educational content that administrators can extend without changing these requirements.

Table 2: Core concepts

| **Domain** | **Core concepts** |
|---|---|
| Investing | Stock; ETF; share; market price; portfolio value; return; diversification; risk and volatility; compound growth; market downturn and recovery |
| Canadian accounts | Tax-Free Savings Account (TFSA); Registered Retirement Savings Plan (RRSP) |
| Fitness | Each supported activity type; active minutes; rest day; streak |

FR39: The system shall provide a concept explanation for each core concept in Table 2.
*Fit criterion:* For every core concept in Table 2, a tester can open a concept explanation of it in the app.
*Traces to:* SR??, SR??, G1

FR40: The system shall let the user open the concept explanation of a core concept from every screen on which that concept appears.

*Fit criterion:* On every screen that shows a core concept, a tester can open its concept explanation without leaving that screen.
*Traces to:* SR**??**, G1, G2

FR41: When the user's portfolio value has changed by more than `FEEDBACK_THRESHOLD` since they last viewed their portfolio, the system shall show feedback naming the holdings that contributed most to the change.
*Fit criterion:* For a test portfolio whose value has changed by more than the threshold, the feedback names the holding whose value changed most.
*Traces to:* SR**??**, G1

FR42: After each trade, the system shall show the user how the trade changed the share of their portfolio value held in each asset.
*Fit criterion:* After a test trade, the feedback shows each holding's share of portfolio value before and after the trade, and these match the values calculated from the holdings.
*Traces to:* SR**??**, SR**??**, G1, G4

FR43: When a single holding is more than `CONCENTRATION_LIMIT` of the user's portfolio value, the system shall show the user an explanation of diversification.
*Fit criterion:* A test portfolio with one holding above `CONCENTRATION_LIMIT` of its value shows the explanation, and the explanation names no asset to buy or sell (Hazard Analysis SAF-9).
*Traces to:* SR**??**, SR**??**, G1, Hazard Analysis SAF-9

FR44: The system shall provide a knowledge check for each core concept in Table 2.
*Fit criterion:* For every core concept in Table 2, a tester can start a knowledge check on it.
*Traces to:* SR**??**, G1

FR45: When the user finishes a knowledge check, the system shall show which of their answers were correct and an explanation of the correct answer to each question they got wrong.
*Fit criterion:* A test user who answers one question wrongly sees that question marked incorrect, together with the explanation of its correct answer.
*Traces to:* SR**??**, G1

FR46: The system shall show the user their learning progress.
*Fit criterion:* After a test user views a concept explanation and passes a knowledge check, both appear in their learning progress.
*Traces to:* SR**??**, G1, G2

FR47: The system shall let administrators add and edit concept explanations and knowledge checks without releasing a new version of the app.
*Fit criterion:* A concept explanation edited by an administrator is shown in its edited form on a test phone whose app has not been updated.
*Traces to:* SR**??**, G1

## 6.5 Social Feed

A user can share only the kinds of update in Table 3. Currency balances and portfolio values are not shareable, so that friends compare effort and learning rather than market results.

Table 3: Kinds of shared update

| **Kind** | **Content** |
|---|---|
| Activity total | The user's total for one activity type over a day or a week |
| Streak | The user's current streak, in days. A rest day does not end a streak (Hazard Analysis SAF-1) |
| Fitness milestone | A fitness milestone the user has reached |
| Learning milestone | A knowledge check the user has passed |

FR48: The system shall let the user send a friend request to another user by their username.
*Fit criterion:* A test user who enters another user's username sends that user a friend request, which the other user sees.
*Traces to:* SR**??**, UC**??**, G3

FR49: The system shall make two users friends only after the user who received the friend request accepts it.
*Fit criterion:* Two test users are not friends while a request between them is pending or after it is declined, and are friends once it is accepted.
*Traces to:* SR**??**, G3

FR50: The system shall let the user decline a friend request.
*Fit criterion:* After a test user declines a request, the two users are not friends and the request is no longer shown to the user who declined it.
*Traces to:* SR**??**, G3

FR51: The system shall let the user remove a friend.
*Fit criterion:* After a test user removes a friend, neither user sees the other's shared updates.
*Traces to:* SR**??**, G3, Hazard Analysis SEC-13

FR52: The system shall let the user share an update of a kind listed in Table 3.
*Fit criterion:* A test user can share one update of each kind in Table 3, and cannot share anything else.
*Traces to:* SR**??**, UC**??**, G3

FR53: The system shall let the user choose which of their friends can see each shared update.
*Fit criterion:* A test update shared with one selected friend appears in that friend's social feed and in no other friend's.
*Traces to:* SR**??**, UC**??**, G3, Hazard Analysis SEC-13

FR54: The system shall share an update only after showing the user the update exactly as their friends will see it.
*Fit criterion:* For every test update, the preview shown before sharing matches what the selected friends see in their social feed.
*Traces to:* SR**??**, UC**??**, G3

FR55: The system shall not include the user's age, height, or weight in any shared update.
*Fit criterion:* No shared update shown to a test user's friends contains their age, height, or weight.
*Traces to:* SR**??**, G3

FR56: The system shall show the user a social feed of the updates their friends have shared with them, newest first.
*Fit criterion:* A test user's social feed contains every update shared with them by a friend and no others, ordered by the time each was shared.
*Traces to:* SR**??**, G3

FR57: The system shall let the user delete an update they have shared.
*Fit criterion:* After a test user deletes a shared update, it no longer appears in any friend's social feed.
*Traces to:* SR**??**, G3, Hazard Analysis SEC-13

FR58: (HA SAF-11) The system shall not rank users against each other by activity, currency or portfolio performance.
*Fit criterion:* No screen shown to a test user orders users by activity, currency or portfolio performance, or shows a user's position relative to others.
*Traces to:* SR**??**, G3

## 6.6 Reminders and Settings

FitFolio sends reminders of the types in Table 4, which are the same reminder types as in the Hazard Analysis.

Table 4: Reminder types

| Type | Example | Feature area |
|---|---|---|
| Activity | Progress toward today's goal, or a streak about to end | Activity and Rewards |
| Milestone | A goal reached, or a streak or currency milestone | Activity and Rewards |
| Portfolio | A notable change in simulated portfolio value | Simulated Market and Portfolio |
| Learning | A new concept explanation or knowledge check is available | Learning and Feedback |
| Social | A friend shared an update or sent a friend request | Social Feed |

FR59: The system shall let the user choose the time periods in which they receive reminders.
*Fit criterion:* A test user sets a reminder period, and it is shown in their settings the next time they open the app.
*Traces to:* SR**??**, UC**??**, G2, Hazard Analysis SAF-6

FR60: (HA SAF-6) The system shall send reminders of the types in Table 4 only during the time periods the user has chosen to receive them.
*Fit criterion:* For a test user who has chosen their reminder periods, no reminder of any type in Table 4 is delivered outside those periods. For a test user who has not chosen any periods, reminders are delivered only during `DEFAULT_REMINDER_PERIOD`.
*Traces to:* SR**??**, UC**??**, G2

FR61: The system shall let the user withdraw consent for FitFolio to read their health data.
*Fit criterion:* After a test user withdraws consent in the app, no further health data is read.
*Traces to:* SR**??**, UC**??**, G3, Hazard Analysis SEC-9

FR62: The system shall show the user which health data types FitFolio has permission to read, and the feature that uses each type.
*Fit criterion:* A test user who has granted health permissions sees each permitted data type listed with the feature that uses it.
*Traces to:* SR**??**, SR**??**, UC**??**, G3, Hazard Analysis SEC-8

# 7 Data Requirements

[The Data Requirements section describes what information the system must manage. It includes the major data entities, their attributes, and the relationships between them. It also includes a data dictionary that defines the meaning of each attribute. —TPLT]

## 7.1 Major Data Entities

[What information must exist for the system to accomplish its goals? —TPLT]

[For each entity, provide a short explanation. —TPLT]

- User
- Song
- Session
- Report

## 7.2 Data Dictionary

[For each entity, define all important attributes. —TPLT]

| Attribute | Type | Description |
|---|---|---|
| user_id | UUID | Unique identifier |
| email | String | User email address |

Table 5: Data Dictionary

## 7.3 Privacy and Data Retention

[Discuss storage, retention, deletion, privacy controls, consent, encryption, and compliance requirements. —TPLT]

TODO

# 8 Non-Functional Requirements

[You may have to add other NFRs. —TPLT]

[List your nonfunctional requirements. You may consider using a fit criterion to make them verifiable. —TPLT]

[The goal is for the nonfunctional requirements to be unambiguous, abstract and verifiable. This isn't easy to show succinctly, so a good strategy may be to give a "high level" view of the requirement, but allow for the details to be covered in the Verification and Validation document. —TPLT]

[An absolute requirement on a quality of the system is rarely needed. For instance, an accuracy of 0.0101 % is likely fine, even if the requirement is for 0.01 % accuracy. Therefore, the emphasis will often be more on describing now well the quality is achieved, through experimentation, and possibly theory, rather than meeting some bar that was defined a priori. —TPLT]

[You do not need an entry for correctness in your NFRs. The purpose of the SRS is to record the requirements that need to be satisfied for correctness. Any statement of correctness would just be redundant. Rather than discuss correctness, you can characterize how far away from the correct (true) solution you are allowed to be. This is discussed under accuracy. —TPLT]

NFR1: **Accuracy** [Characterize the accuracy by giving the context/use for the software. Maybe something like, "The accuracy of the computed solutions should meet the level needed for <engineering or scientific application>. The level of accuracy achieved by FitFolio shall be described following the procedure given in Section X of the Verification and Validation Plan." If you do this provide a link to the VnV plan. —TPLT]

NFR2: **Usability** [Characterize the usability by giving the context/use for the software. You should likely reference the user characteristics section. The level of usability achieved by the software shall be described following the procedure given in Section X of the Verification and Validation Plan. Include a link to the VnV plan. —TPLT]

NFR3: **Maintainability** [The effort required to make any of the likely changes listed for FitFolio should be less than FRACTION of the original development time. FRACTION is then a symbolic constant that can be defined at the end of the report. —TPLT]

NFR4: **Portability** [This NFR is easier to write than the others. The systems that FitFolio should run on should be listed here. When possible the specific versions of the potential operating environments should be given. To make the NFR verifiable a statement could be made that the tests from a given section of the VnV plan can be successfully run on all of the possible operating environments. —TPLT]

- Other NFRs that might be discussed include verifiability, understandability, reusability, performance, reliability and availability, security, and accessibility. You may have to add other NFRs.

# 9 AI and Machine Learning Requirements

[Refer to Dr. Mosser's guest lecture on AI requirements. —TPLT]
[Include only when applicable. —TPLT]

## 9.1 Model Inputs

TODO

## 9.2 Model Outputs

TODO

### 9.3 Evaluation Metrics

[Accuracy, Precision, Recall, F1, BLEU, WER, etc. —TPLT]

TODO

### 9.4 Known Limitations

[Explicitly document expected failure modes. —TPLT]

TODO

## 10 Requirements Traceability Matrices

[The Requirements Traceability Matrix (RTM) is a table that shows the relationship between different types of requirements. It is used to ensure that all requirements are traceable and that no requirements are missed. The typical flow is: Stakeholder Need -¿ Use Case -¿ Requirement —TPLT]

[There is also a traceability matrix between functional and nonfunctional requirements. The purpose of some FRs is to satisfy NFRs. Therefore, it is important to show the relationship between the two. —TPLT]

[Further traceability to test cases and modules will be given in other documents. —TPLT]

| Stakeholder Req | Use Case | Functional Req |
|---|---|---|
| SR1 | UC1 | FR13 |

Table 6: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

| Functional Req | Nonfunctional Req |
|---|---|
| FR13 | NFR1 |

Table 7: Traceability Between Functional and Nonfunctional Requirements

## 11 Rationale

[Provide a rationale for the decisions made in the documentation. Rationale should be provided for scope decisions, functional requirements, nonfunctioanl requirements, data requirements, and AI requirements. You don't need a rationale for everything, ust the major decisions that are not obvious. You should have traceability to the entities in the document that you are referencing. —TPLT]

# 12 Likely Changes

The following changes are identified in the project documents. They describe parts of FitFolio that may change as the team tests its planned integrations or adds features already listed as stretch goals.

LC1: **Activity-data updates**

When FitFolio updates activity data may change because of restrictions imposed by the phone's health platform. The Development Plan identifies updating activity data when the user opens the app as an alternative if background updates cannot work reliably. This would affect when activity data is retrieved in UC1 (Record Daily Activity) and becomes available for awarding currency under FR13. Any change must still follow the health-platform access rules in SC3.

LC2: **Wearable device support**

FitFolio may be extended to support wearable devices, such as the Apple Watch and other compatible fitness trackers. Wearable integration is already listed as a stretch goal in the Problem Statement and Goals. Adding this support could provide additional activity data, including workout duration and calories burned. This would affect the Activity Record entity in Table **??** and its fields in Table 5. Direct access to a wearable, rather than reading its data through the phone's health platform, would also require revising SC2. Wearables would remain optional, as required by SR**??**.

LC3: **Market-data access**

The market-data provider, price-update frequency, and selection of supported assets may change if the chosen service's request limits or costs prevent the planned use. The Development Plan identifies switching providers, reducing the update frequency, and limiting the asset list as possible responses. These changes would affect the provider choice in SC4, the prices obtained under FR26, and the available assets represented by the Market Data entity in Table **??**. Changes to the update frequency would also need to account for the price-age display in FR27 and the restriction on trading with stale prices in FR33. Any reduction below the price-freshness or asset-coverage targets in Section 2.2 would require an agreed revision to those targets and the affected requirements.

LC4: **Historical market scenarios**

The number and selection of historical market scenarios may change depending on the availability and cost of historical price data. The Development Plan identifies offering fewer scenarios or using a selected set of stored historical datasets as possible responses if the required data is difficult to obtain. This would affect the past market conditions available under SR**??** and the historical prices represented by the Market Data entity in Table **??** and its `price_history` field in Table 5.

LC5: **Social challenges and leaderboards**

FitFolio may be extended to include fitness challenges and opt-in leaderboards. Social challenges and leaderboards are listed as a stretch goal in the Problem Statement and Goals, and the Hazard Analysis identifies opt-in leaderboards as a possible future feature. The current version prohibits ranking users by activity, currency, or portfolio performance under FR58 and Hazard Analysis SAF-11. Adding leaderboards would therefore require an agreed revision to those requirements and a review of the risk of harmful comparison identified in SR**??**. Adding comparisons of portfolio statistics would also require revisiting the allowed updates in FR52 and Table 3. Users would still choose whether to participate and which information to share, consistent with SR**??** and FR53.

# 13 Unlikely Changes

The following parts of FitFolio are expected to stay the same because they support the project's purpose and goals.

ULC1: **Virtual currency and simulated investing**

FitFolio is expected to continue using virtual currency with no real-world monetary value. Adding real-money trading or allowing users to exchange earned currency for cash is unlikely. The Problem Statement and Goals describes an app where beginners can practise investing without risking real money or needing to link bank or brokerage accounts. Moving to real money would change this purpose.

ULC2: **Earning rewards through physical activity**

Physical activity is expected to remain the way users earn virtual currency in FitFolio. Removing activity-based rewards is unlikely because earning currency through exercise and using it in the simulated market is the project's central idea. This connection supports the goal of helping users build fitness habits while learning about investing.

ULC3: **User control over social sharing**

Users are expected to keep control over which activity and portfolio information they share. Removing this control is unlikely because the Problem Statement and Goals includes motivating users through friends while protecting their privacy. It also states that raw health data is shared only when the user chooses to share it. Any changes to the social features should preserve this choice.

# 14 Appendices

## 14.1 Appendix A: Additional Materials

TODO

## 14.2 Appendix B: Development Plan

[This section is optional. It is used to explain the plan for developing the software. In particular, this section gives a list of the order in which the requirements will be implemented. In the context of a course this is where you can indicate which requirements will be implemented as part of the course, and which will be "faked" as future work. This section can be organized as a prioritized list of requirements, or it could show the requirements that will be implemented for "phase 1", "phase 2", etc. —TPLT]

## 14.3 Appendix C: Values of Auxiliary Constants

[Show the values of the symbolic parameters introduced in the report. —TPLT]

[The definition of the requirements will likely call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance. —TPLT]

[The value of FRACTION, for the Maintainability NFR would be given here. —TPLT]

Table 8 gives the value of each symbolic constant used in this document and in the Hazard Analysis (HA).

| Constant | Value | Used in |
|---|---|---|
| `MIN_USER_AGE` | 18 | FR2 |
| `MIN_ANDROID_VERSION` | 17 | SC1 |
| `MIN_IOS_VERSION` | 27 | SC1 |
| `MARKET_DATA_PROVIDER` | Finnhub (Development Plan) | SC4 |
| `MIN_ACTIVITY_TYPES` | 5 (G5) | FR12 |
| `EARNING_RATES` | 1 per 100 steps. Workouts without a step count, such as cycling or swimming, earn 1 per minute | FR13 |
| `MAX_DAILY_CURRENCY` | 300 | FR13, FR16, HA SAF-7 |
| `MIN_TRADABLE_ASSETS` | 50 (G4) | FR25 |
| `QUOTE_STALENESS_LIMIT` | 15 minutes (G4) | FR27, FR33, HA SEC-2, SEC-3 |
| `FEEDBACK_THRESHOLD` | 5% | FR41 |
| `CONCENTRATION_LIMIT` | 50% | FR43 |

| Constant | Value | Used in |
|---|---|---|
| DEFAULT_REMINDER_PERIOD | 9:00 to 20:00 in the user's time zone | FR60, HA SAF-6 |
| REST_DAYS_PER_PERIOD | 2 | FR21, HA SAF-1 |
| STREAK_PERIOD | 7 days | FR21, HA SAF-1 |
| GOAL_REVIEW_PERIOD | 7 | FR19, HA SAF-4 |
| MAX_GOAL_INCREASE | 10% | HA SAF-5 |
| MILESTONES | Streaks of 7, 30 and 100 days, and 1,000, 10,000 and 100,000 currency earned in total | FR23 |
| MAX_DAILY_STEPS | 150,000 | FR15, HA SEC-1 |
| MAX_WORKOUT_DURATION | 12 hours | FR15, HA SEC-1 |
| SESSION_TIMEOUT | 14 days | HA SEC-17 |
| DATA_DELETION_PERIOD | 30 days | HA SEC-18 |

Table 8: Values of auxiliary constants

## 14.4 Appendix D: Generative AI Disclosure

[Provide full transparency on how generative AI was used in the creation of this document. This includes any text, figures, or tables that were generated by AI. If any skills were created for AI agents to review the documentation, those should be disclosed here. —TPLT]

## 14.5 Appendix E: Reflection

[This isn't part of the SRS, but it is part of the course. —TPLT]

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

**2. What pain points did you experience during this deliverable, and how did you resolve them?**

**Davis:** The main pain point for me was how many design decisions the SRS involved, and how many of them crossed between sections owned by different people. Farid and I split the functional requirements, Angad wrote the use cases and traceability matrices, and Preston wrote the data dictionary, but all of these depend on the same choices, such as what a user can share with friends and whether activity data leaves the phone as individual records or daily totals. Because we each worked in our own branch and only compared our work near the deadline, we found these mismatches late. Some had a ripple effect, for example, we had used different words for the same thing, such as “post” and “shared update”, one of the shared tables contradicted a Hazard Analysis requirement on how rest days affect a streak, and no one had filled in the appendix of constants that many of our requirements relied on. As a result, we spent much of the last few days aligning our sections and resolving merge conflicts instead of improving the content. I resolved this by reviewing my teammates’ open pull requests against my sections, adopting the terms Farid defined in Section 1.4, filling in the appendix of constants, and listing what I still needed from each person in my pull request. Next time, I want to communicate with the team earlier and more often. That means agreeing on shared identifiers, terms and the big design decisions at the start of a deliverable, recording each decision in the related GitHub issue as soon as it is made, and merging smaller pieces of work sooner, so that conflicts show up while they are still cheaper to fix.